

Hadoop streaming with R tutorial

In this tutorial we will work through steps to run the word count example with mappers and reducers written in R and show you how to use hadoop streaming to perform the ordinary map-reduce.

You need a machine with hadoop and R installed on it. If you use the Ubuntu image that we provided you need to install R via this command:

```
sudo apt-get install r-base
```

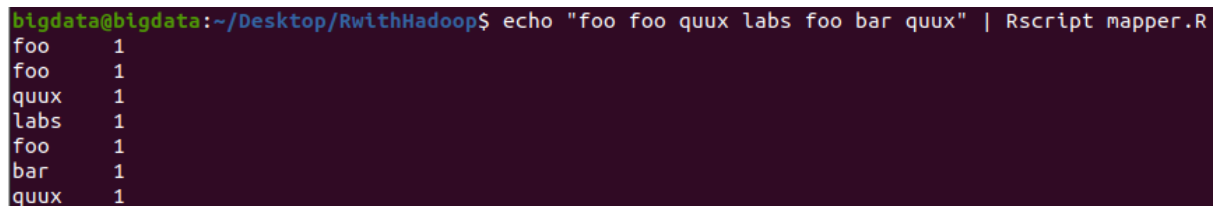
Steps to run a map-reduce program in R with Hadoop:

1- Write your map and reduce logic in two separate files (e.g. mapper.R & reducer.R)

2- Both mapper and reducer reads input via stdin and output via stdout to work with Hadoop streaming → refer to mapper.R and reducer.R

3- Let's test locally the map logic

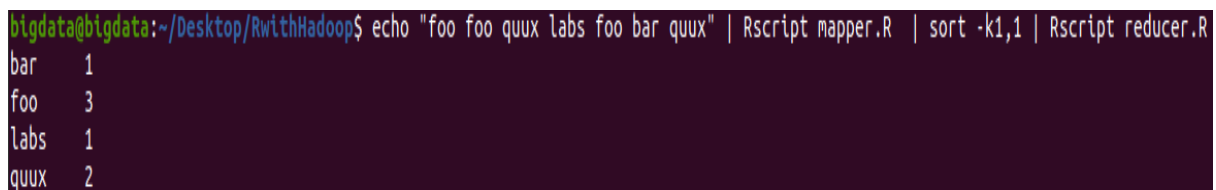
```
echo "foo foo quux labs foo bar quux" | Rscript mapper.R
```



```
bigdata@bigdata:~/Desktop/RwithHadoop$ echo "foo foo quux labs foo bar quux" | Rscript mapper.R
foo      1
foo      1
quux     1
labs     1
foo      1
bar      1
quux     1
```

4- Let's test locally the reduce logic

```
echo "foo foo quux labs foo bar quux" | Rscript mapper.R | sort -k1,1 | Rscript reducer.R
```



```
bigdata@bigdata:~/Desktop/RwithHadoop$ echo "foo foo quux labs foo bar quux" | Rscript mapper.R | sort -k1,1 | Rscript reducer.R
bar      1
foo      3
labs     1
quux     2
```

At this point we are sure that map and reduce logic works as expected, Let's move to Hadoop

5- Start hadoop

```
start-all.sh
```

6- Create directories on HDFS for this tutorial

```
hdfs dfs -mkdir /RwithHadoop
```

hdfs dfs -mkdir /RwithHadoop/Input

7- move input.txt from local FS to HDFS (make sure that the terminal current directory is where input.txt exists)

hdfs dfs -put input.txt /RwithHadoop/Input

8- Now, we need to run Hadoop streaming with mapper.R and reducer.R and for this to work we need to find hadoop-streaming.jar via this command:

sudo find / -name 'hadoop-streaming*.jar'

This will ask you for the password (if you use our machine the password is: 123456)

```
bigdata@bigdata:~/Desktop/RwithHadoop$ sudo find / -name 'hadoop-streaming*.jar'
[sudo] password for bigdata:
find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.0.jar
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.0-test-sources.jar
/usr/local/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.0-sources.jar
```

The path that we need here is:

/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.0.jar

Where 3.3.0 is the hadoop version.

This jar file is the hadoop streaming where it can take external mappers and reducers with your favourable language (R, Python, ...)

9- Run the map reduce job via this command:

hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.0.jar -file /home/bigdata/Desktop/RwithHadoop/mapper.R -mapper /home/bigdata/Desktop/RwithHadoop/mapper.R -file /home/bigdata/Desktop/RwithHadoop/reducer.R -reducer /home/bigdata/Desktop/RwithHadoop/reducer.R -input /RwithHadoop/Input -output /RwithHadoop/Output

10- Print the map reduce output:

hdfs dfs -cat /RwithHadoop/Output/*

```
bigdata@bigdata:~/Desktop/RwithHadoop$ hdfs dfs -cat /RwithHadoop/Output/*
2022-03-14 12:59:11,254 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
ahmed 2
galal 1
hello 2
mohamed 2
omar 3
samir 1
```